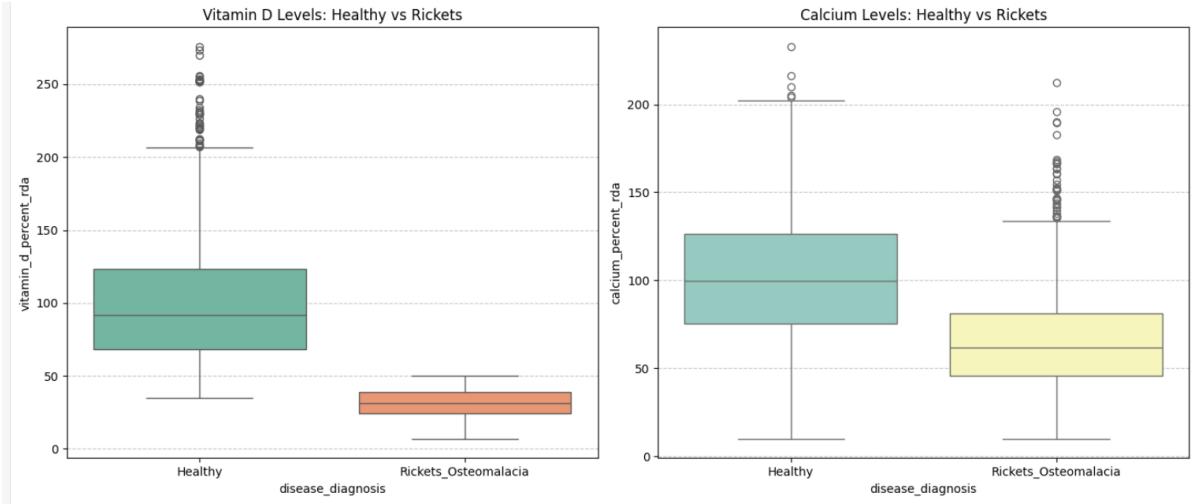
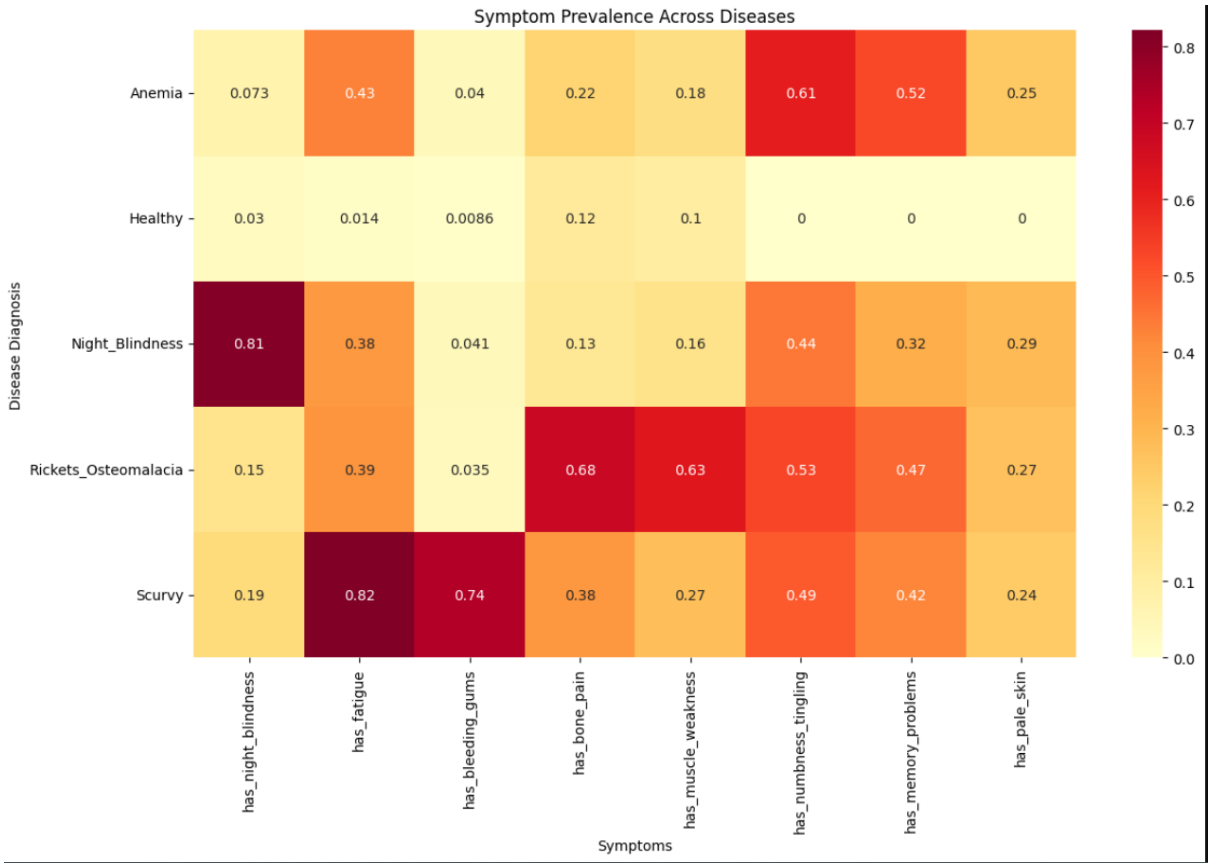
Exploratory Data Analysis

1. What is the frequency of each *disease_diagnosis* in the dataset?
2. What are the average haemoglobin, iron, and serum folate levels for Anaemia vs Healthy individuals?
3. How do vitamin D and calcium levels differ between Rickets/Osteomalacia and Healthy individuals?
4. Which symptoms are most common in each disease?
5. Which diet type has the highest number of Anemia cases?
6. How does sun exposure affect Vitamin D levels?
7. Is there an association between alcohol consumption and disease diagnosis?
8. Does smoking status affect Vitamin C levels?
9. Does disease distribution differ between males and females?
10. Does income level affect disease occurrence?
11. Which clinical markers show the highest variation across diseases?
12. What BMI range is most common among healthy individuals?
13. Do symptoms like bone pain and fatigue occur together in Rickets patients?
14. How does symptoms count vary across different diseases?
15. Which vitamin deficiency markers are strongest predictors for each disease category?
16. Do hemoglobin levels significantly differ across disease categories?


Vitamin D Levels: Healthy vs Rickets
vitamin_d_percent_rda
0
50
100
150
200
250
Healthy
Rickets_Osteomalacia
disease_diagnosis
Calcium Levels: Healthy vs Rickets
calcium_percent_rda
0
50
100
150
200
Healthy
Rickets_Osteomalacia
disease_diagnosis


Symptom Prevalence Across Diseases
Disease Diagnosis
Anemia 0.073 0.43 0.04 0.22 0.18 0.61 0.52 0.25
Healthy 0.03 0.014 0.0086 0.12 0.1 0 0 0
Night_Blindness 0.81 0.38 0.041 0.13 0.16 0.44 0.32 0.29
Rickets_Osteomalacia 0.15 0.39 0.035 0.68 0.63 0.53 0.47 0.27
Scurvy 0.19 0.82 0.74 0.38 0.27 0.49 0.42 0.24
has_night_blindness
has_fatigue
has_bleeding_gums
has_bone_pain
has_muscle_weakness
has_numbness_tingling
has_memory_problems
has_pale_skin
Symptoms
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8